

Basic Possibilistic Logic: Examples and Visualization

Seminar Paper

submitted by

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Abstract

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1 Introduction

The goal of this paper is to introduce Basic Possibilistic Logic with the use of examples and visualization. Possibilistic logic is a weighted logic that handles uncertainty in a qualitative way by adding certainty to classical logic formulas. Its strength lies in modeling uncertainty and its ability to deal with inconsistency. It has been developed for about thirty years. It still remains very close to classical logic. In this paper we will only focus on basic Possibilistic Logic.

This work is structured as follows. In the Section 2 Basic Possibilistic Logic is explained with its Preliminaries for understanding the syntax and semantics of the logic. The Section 3 shows the use of Basic Possibilistic Logic with multiple examples. Section 4 explains Basic Possibilistic Logic with graphical explanations. The last Section 5 discusses the results of the previous sections.

2 Fundamentals of Basic Possibilistic Logic

As a main source the overview chapter: Possibilistic Logic - An Overview from the authors Dubois and Prade [1] is used.

2.1 Introduction to Possibility Theory

2.1.1 Possibility Distribution

A possibility distribution is a mapping π from U to a totally ordered scale S , with top denoted by 1 and bottom by 0:

$$\pi : U \rightarrow S,$$

where U is the set of states. $\pi(u)$ can be understood as the degree of possibility of the state u . The function π represents the state of knowledge of an agent and distinguishes what is plausible from what is less plausible. The state u is impossible, if $\pi(u)$ equals 0. The state is fully possible, if $\pi(u)$ equals 1.

A distribution π is at least as specific as π' if

$$\pi(u) \leq \pi'(u), \quad \forall u \in U.$$

An agent has complete knowledge if the possibility of one state is 1 and the possibility of all other states is 0. Complete ignorance is given if the possibility of all states is 1. The Principle of minimal specificity states that any hypothesis not known to be impossible cannot be ruled out. It is a minimal commitment, cautious, information principle.

2.1.2 Possibility and Necessity Functions

If we want to know if the event A occurs we can respond in possibility theory with two measures calculated from the possibility distribution to A . A is a subset of states. The two measures are the possibility and the necessity of an event.

The Possibility is defined as $\Pi(A) = \sup_{u \in A} \pi(u)$. It is the maximum of possibility of all the states of A . The Possibility evaluates the extent to which A is consistent with π . Possibility measures satisfy the characteristic maxitivity property $\Pi(A \cup B) = \max(\Pi(A), \Pi(B))$.

The Necessity is defined as $N(A) = \inf_{u \notin A} (1 - \pi(u))$. It is the minimum of the inversed possibility value of all states which are not in A . The Necessity evaluates the extent to which A is certainly implied by π . Necessity measures satisfy the characteristic minimization property $N(A \cap B) = \min(N(A), N(B))$. The minimization property expresses that to be certain about A and B , one has to be certain about A and to be certain about B .

The duality of the two measures is expressed as $N(A) = 1 - \Pi(A^c)$, where A^c is the complement of A .

2.2 Formal foundations of Basic Possibilistic Logic

A basic possibilistic logic formula is represented as:

$$(a, \alpha)$$

where a is a classical logic formula (e.g., $p \rightarrow q$) and α is the certainty level (minimum necessity degree of a). It can be viewed as a lower bound of a necessity measure. The

pair (a, α) is understood as $N(a) > \alpha$. We can interpret that the formula a is at least certain to degree α . The higher α , the stronger the belief in a . For example $(b \rightarrow f, 0.9)$ means that birds fly with certainty 0.9.

2.2.1 Axioms and Inference Rules

Syntax

Semantics

Differences from classical logic

Relation to Artificial Intelligence

3 Analysis and Development of Examples

3.1 Elaboration on existing examples from the literature

The „Tweety penguin problem “

Other relevant examples from Dubois’ works

3.2 Extension of existing examples

Detailed discussion and interpretation

3.3 Development of new examples

Motivation for the new examples

Presentation and analysis

4 Visualization in Possibilistic Logic

4.1 The need for and benefits of visualization

4.2 Approaches to visualization

Graphical representations of possibility and necessity

Diagrams illustrating inference processes

4.3 Implementation and evaluation

5 Discussion and Conclusion

5.1 Summary of findings

5.2 Limitations of the study

5.3 Outlook on future research

6 Example Figures and Tables

Figure 1 shows an example of an abstract argumentation framework.

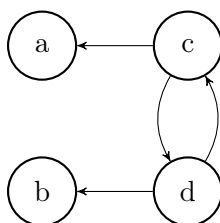


Figure 1: An abstract argumentation framework.

1	2	3
4	5	6

Table 1: Table caption.

6.1 Encoding Test

Städte, Länder, Flüsse

References

- [1] Didier Dubois and Henri Prade. Possibilistic logic — an overview. In Jörg H. Siekmann, editor, *Computational Logic*, volume 9 of *Handbook of the History of Logic*, pages 283–342. North-Holland, 2014.

Statement

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